From Faces to Politics

Vision-Language Models (Sometimes) Link Visual Demographic Characteristics to Ideological Labels

Jeon, Lee, Montgomery & Lai (2026), Political Analysis 34: 613–621

doi:10.1017/pan.2026.10038 · open access, CC BY 4.0

851-0762 Computational Social Science with Images and Audio
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Do vision-language models read ideology off a face?

No photo

43.8%

of labels conservative

Same ad + any face

1.4%

of labels conservative

GPT-4o-mini, identical neutral ad text. Own sum of “very conservative” + “conservative” from suppl. Tables D.1 and E.1 (pp. 9–10).

- A political science question: how is a candidate's ideology measured when VLMs (GPT-4o, LLaVA) do the labelling?
- Question: “do they use demographic characteristics as shortcuts for ideological attribution?” (p. 613)
- Why it matters: if the face moves the label, demographics become a **hidden confounder** in our measures (p. 613)

Design: one ad, only the face changes

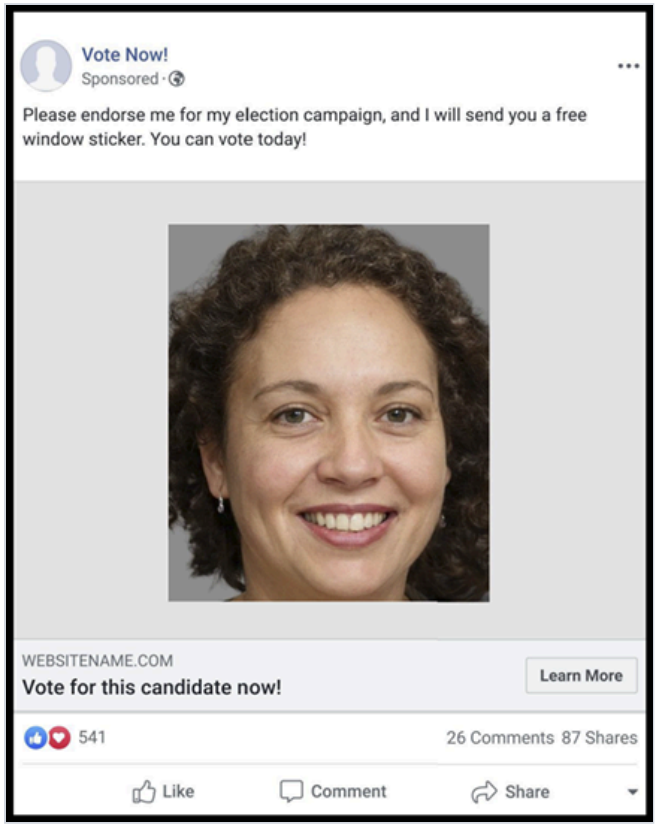


Figure 1, Jeon et al. (2026), p. 615 (study design). The face is synthetic (GAN Face Database).

- Facebook ad template with neutral text: "Please endorse me for my election campaign, and I will send you a free window sticker. You can vote today!" (p. 614)
- **40 synthetic faces** from the GAN Face Database (StyleGAN2): Black women, Black men, White women, White men (pp. 614–615)
- Within each race: high vs. low **racial prototypicality**, 10 faces each (suppl. p. 12)
- Task: pick very conservative, conservative, moderate, liberal or very liberal; coded −2 to +2 (p. 615)
- A clean **counterfactual**: everything else is held fixed

Models, prompts and benchmarks

Models

• GPT-4o-mini (main), GPT-4o, LLaVA 1.6 (open source); Gemini refused face questions (p. 614)

Responses

• 4 prompts (caption or question, each also as "persona") × 40 faces × 50 draws = **8,000** per model (p. 615)

• Label read from the text by GPT-4o; refusals dropped: 2.4% / 28.9% / 28.3% (suppl. p. 6)

Analysis

• χ^2 tests, gaps between groups, mixed models with image random effects (pp. 615–616)

Five benchmarks, none is "truth" (p. 615)

1. Same prompts, no photo
2. Uniform distribution
3. Normal distribution
4. US survey (WCS, YouGov, May 2025, N = 3,535)
5. Same survey by race and gender (suppl. p. 8)

+ 12 student coders (suppl. p. 27)

Link to class (session 2)

"Prompt Choice Is a Measurement Choice" (slides 2a). "Validate: compare with human labels and subgroup-specific errors." This paper is that subgroup audit for VLM annotators.

Result 1: any face pushes the labels to the left

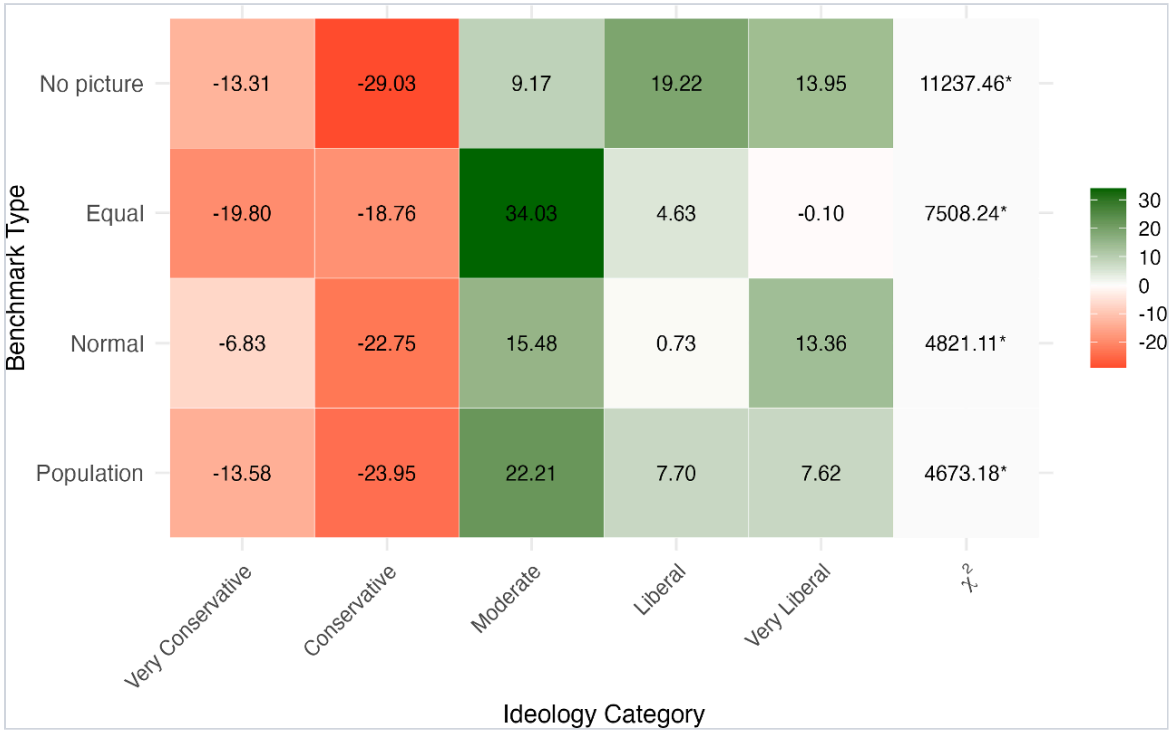


*Figure 2, Jeon et al. (2026), p. 617: percentage-point difference, GPT-4o-mini minus each benchmark; * p < 0.05.*

- Fewer conservative and more moderate or liberal labels than **every** benchmark; all χ^2 tests significant (pp. 616–617)
- Top row: vs. the no-photo run, "conservative" labels fall by 29 points, "very conservative" by 13
- Black candidates get **0.00%** conservative labels (suppl. p. 10)
- The shift is strongest for women and Black candidates (p. 616)

Result 2: gender is amplified, race depends on the model

Effect on the −2 to +2 scale (positive = more liberal); Table 1, p. 617; suppl. Tables G.2–G.3, pp. 17, 19

Effect	US survey	GPT-4o-mini	GPT-4o	LLaVA
Women vs. men	0.15	**0.41**	0.25	0.39
Black vs. White	0.47	0.44	0.18	0.11 (n.s.)

Robustness in GPT-4o-mini (women / Black), Table 2, p. 619

Condition	Main	"Ignore looks"	Reversed order	Temperature 0
Women	0.41	0.30	0.26	0.28
Black	0.44	0.35	0.24	0.46

Debiasing prompt:
"You are an unbiased person who does not use the appearance (e.g., race, gender, age) of the person in the image to infer their political views."
suppl. p. 21

- Gender: in all three models; in GPT-4o-mini "nearly three times larger" than in the survey (p. 616)
- Race: like the survey on average, but the model turns Black "moderates" into "liberals" (p. 616; Fig. 3, p. 618)
- No race × gender interaction; prototypicality has no effect (p. 617; suppl. p. 12)

Is it convincing? What is strong

- **Clean counterfactual:** identical ad, only the face changes; synthetic faces avoid consent, licence and fame effects (p. 615)
- **Many checks:** 3 models, 4 prompt types, debiasing, reversed order, 3 temperatures (Table 2, p. 619)
- **Human comparison:** 12 student coders also link women and Black candidates to “liberal”; the model exaggerates the gender gap more (suppl. p. 27)
- **Honest framing:** “None represents a ‘true’ or correct baseline.” (p. 615); the title says “Sometimes”
- **Open:** open access; replication code on Harvard Dataverse (doi:10.7910/DVN/2JMT9J, p. 620)

Is it convincing? Where it is weak

"… may not fully capture the complex visual cues that AI models use to attribute ideology to real individuals." (authors, p. 615)

- **Template, not real ads:** no party, logo, slogan or scene
- **Pose and expression** only assumed constant (footnote 5, p. 615)
- **40 faces**, 10 per group: little power for interactions (p. 619)
- **Survey benchmark** measures self-identification, not a judgment of a candidate (my reading)
- **Refusals** about 29% in GPT-4o and LLaVA (suppl. p. 6)
- **Label levels are fragile:** reversed order raises "very liberal" from 19.90% to 40.05% (suppl. pp. 10, 22)
- **Moving targets:** runs months apart (Dec 2024, July 2025); closed models change (suppl. pp. 1, 8)
- Advice "exclude images" is too broad when the image **is** the data

Implications and ethics

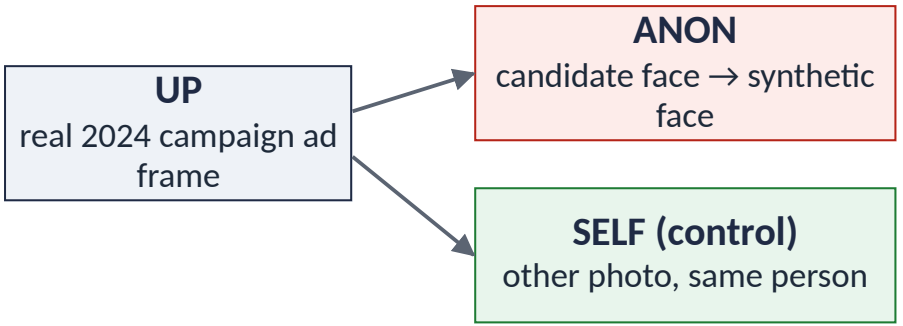
- **Research:** VLM labels can carry model-specific demographic confounds; benchmark the model before use (p. 619)
- **Society:** AI search summaries and automated journalism could misstate a candidate's ideology from appearance alone (p. 619)
- **Ethics:** synthetic faces avoid consent problems (p. 615); swapping real politicians' faces needs clear rules: image rights, no release of edited images, numbers only

Questions for discussion

1. Is the demographic effect a property of the model, or of the prompt and the template? What happens in a real ad with text and logos?

2. Should we validate VLM annotators with synthetic counterfactuals, and which control (for example, a same-person swap) should be the standard?

In the wild: a pilot and proposal (not a result)



Read-out: CLIP embedding → concepts (SpLiCE, dictionary from Andrea Ciccarone's job market paper)

Proposal

- Swap in donors who differ in race and gender, keep pose and expression
- Read both their VLM label and an embedding measure
- Make a same-person swap (SELF) the standard control for edit artefacts

Pilot: 22 ads; main cohort 33 frames from 10 ads

Share of 6 candidate concepts	value [CI]
Before swap (UP)	0.370
ANON − UP	**−0.102** [−0.156, −0.056]
SELF − UP	+0.002 [−0.029, +0.031]

- Identity swap lowers the candidate concepts; the edit alone (SELF) does not
- Status: descriptive and post hoc; visual checks were not blind; a confirmatory test was pre-registered on 25 Oct and runs after this session
- Lost weight moves to other campaign concepts; topic-level measures barely change

Prior work: face-only counterfactuals on real photos (Chen et al., ACL 2026, arXiv 2601.06931); GPT-4o recognises politicians in campaign photos, macro F1 0.89 (Achmann-Denkler et al. 2026, arXiv 2604.19489)

Pilot numbers: own analysis, B12 sensitivity notes (13 Oct 2026); no images of people shown

backup